

Hypercalcaemia & Thyroid/Parathyroid Review (Dr Sarmad)

The Big Picture

Hypercalcaemia = serum Ca above normal (**8.5–10.2 mg/dL**). Most are mild and asymptomatic; the danger is **hypercalcaemic crisis (Ca >14 mg/dL)** — a medical emergency. Treatment rests on three pillars: **rehydrate, force renal excretion, block bone release**. This lecture also reviews high-yield thyroid/parathyroid surgical facts via MCQs.

Hypercalcaemia Basics

- Normal serum Ca **8.5–10.2 mg/dL**; found in up to **5% of hospitalised patients**.
- Most asymptomatic, mild-moderate (**12–14 mg/dL**) — treat underlying cause + diet.

Hypercalcaemic Crisis

- **Ca >14 mg/dL** with symptoms → needs aggressive management.
- **Presentation:** acute confusion, abdominal pain, vomiting, **dehydration, anuria**.
- **ECG:** prolonged PR + **shortened QT** → risk of lethal arrhythmia. **Ca >18 mg/dL** → **coma, cardiac arrest**.

Treatment (3 principles)

1. ↑ **renal excretion** — **aggressive rehydration** (normal saline 200–500 mL/h to keep urine >100 mL/h); then **loop diuretic (furosemide)** once volume restored. Most normalise with this alone.
2. ↓ **skeletal release** (if malignancy + Ca >12):
 - **Bisphosphonates** — inhibit osteoclasts (IV, poor gut absorption).
 - **Calcitonin** — fast but short-acting; **bridge** until bisphosphonate works.
 - **Glucocorticoids (prednisolone)** — enhance calcitonin, ↑ calciuresis, ↓ gut absorption; useful in **vitamin D excess**.
3. **Treat the underlying cause.**

Thyroid/Parathyroid High-Yield Review (from MCQs)

- **Thyroglossal duct cyst** — midline congenital remnant of thyroid descent; remove with **Sistrunk operation** (+ central hyoid).
- **Thyroid blood supply** — superior thyroid (from **external carotid**), inferior thyroid (from **thyrocervical trunk**), thyroid ima (from aorta in 1–4%).
- **Recurrent laryngeal nerve injury** → hoarseness (unilateral) / airway obstruction (bilateral). **Superior laryngeal nerve (external branch)** → can't "**hit high notes**" (cricothyroid).
- **TSH** directly triggers T3/T4 release (TRH → TSH); thyroglobulin tyrosines iodinated → T3/T4.
- **Total/near-total thyroidectomy** preferred for Graves when: RAI contraindicated, **goitre >80 g/airway risk**, poor compliance.
- **FNAB "suspicious for malignancy"** → **lobectomy** (60–75% turn out malignant).
- **Papillary thyroid cancer** — 80% of thyroid cancers; predominant in children + **radiation-exposed (Chernobyl)**.
- **Follicular cancer (2 cm)** → **total thyroidectomy**; recurrence treated with **radioiodine**.
- **Medullary thyroid cancer** → screen for **MEN2** (pheochromocytoma + hyperparathyroidism in 2A; ganglioneuromas/Marfanoid in 2B); spontaneous 75%, familial 25%.

- **Post-thyroidectomy emergency needing re-op = wound haematoma** (compresses trachea → dyspnoea; open at bedside). Hypocalcaemia → IV calcium; bilateral vocal cord palsy → reintubate.

Walk-Away Points

- Normal Ca **8.5–10.2**; crisis **>14**, coma **>18**; ECG = **short QT**.
- Treat crisis: **saline rehydration** → **furosemide** → **bisphosphonate** (calcitonin as bridge).
- **Thyroglossal cyst** → **Sistrunk**; superior thyroid artery from external carotid.
- **Haematoma = the thyroidectomy emergency**; medullary cancer → think MEN2.

Hypercalcaemia & Thyroid/Parathyroid Review — Revision Table

Hypercalcaemia	Value
Normal Ca	8.5–10.2 mg/dL
Mild-moderate	12–14 mg/dL (asymptomatic)
Crisis	>14 mg/dL (aggressive Rx)
Coma/arrest	>18 mg/dL
ECG	Long PR, short QT
Crisis presentation	Confusion, abdominal pain, vomiting, dehydration, anuria
Treatment pillar	Agent
↑ renal excretion	Saline 200–500mL/h then furosemide
↓ bone release	Bisphosphonate (osteoclast inhibit)
Bridge	Calcitonin (fast, short)
Adjunct	Glucocorticoid (vit D excess)
Thyroid review	Fact
Thyroglossal cyst	Midline; Sistrunk operation
Superior thyroid artery	From external carotid
Inferior thyroid artery	From thyrocervical trunk
RLN injury	Hoarse (uni) / airway (bilateral)
Superior laryngeal n.	Can't hit high notes
TSH	Directly releases T3/T4
Thyroid cancer	Management
FNAB "suspicious"	Lobectomy
Papillary	80%, radiation/Chernobyl, children
Follicular	Total thyroidectomy + radioiodine
Medullary	Screen MEN2 (pheo + hyperparathyroid)
Post-thyroidectomy emergency	Wound haematoma → bedside open + re-op

Bold = high-yield.